

William Kimball

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TECHNICAL SKILLS

Languages: Python, Java, C, Swift, Dart, SQL, PostgreSQL, JavaScript, HTML/CSS, R
Frameworks: React, Node.js, Flutter, Django, REST API, Dash
Developer Tools: Git, Google Cloud Platform (BigQuery), Databricks, AWS, Heroku
Libraries: Scikit-Learn, TensorFlow, Pandas, NumPy, Transformers, spaCy, NetworkX, Bootstrap, Qiskit
ML/AI: LLMs, Agentic AI, Retrieval-Augmented Generation (RAG), Vector Search, NLP, Time-Series, Ensemble Methods (XGBoost, RF), Sentiment Analysis, Feature Engineering

EXPERIENCE

Data Engineer June 2026 – Present
Versetal Information Systems Quincy, MA

- Developed an **LLM-powered** internal chatbot using **retrieval-augmented generation (RAG)** to answer employee support requests from historical ticketing data and enterprise knowledge bases.
- Engineered scalable semantic search pipelines by vectorizing ticket records and conversation histories in **Google Cloud Platform (BigQuery)**, enabling high-performance retrieval over enterprise support data.
- Built **agentic AI** workflows using **Anthropic Claude** to intelligently route requests between vector search and **SQL-based** retrieval while automating ticket triage through contextual reasoning and knowledge base retrieval.

Founder & AI Engineer Jan 2026 – Present
Process Trace AI Boston, MA

- Built a full-stack **AI-powered** pharmaceutical investigation assistant using **FastAPI**, **React**, and **DuckDB** to synthesize manufacturing deviations, CAPAs, and quality records into review-ready investigations.
- Designed an **agentic AI** platform using **retrieval-augmented generation (RAG)**, ontology-driven validation, and evidence traceability to deliver explainable, source-backed investigation recommendations.
- Led product strategy, customer discovery, and technical development with pharmaceutical manufacturing and quality leaders, translating customer feedback into platform capabilities and workflow improvements.

Data Scientist May 2025 – Dec 2025
Takeda Pharmaceuticals Cambridge, MA

- Developed and optimized **Random Forest** and **large language models (LLMs)** using **Scikit-Learn** and **NLP** to predict hazardous chemical regulatory actions and classify regulatory text for enterprise compliance.
- Processed material qualification data across 30+ manufacturing sites (10,000+ records) using **Python (Pandas)**, **SQL**, and **Databricks**, performing data cleaning, feature engineering, and validation for downstream analytics.
- Built interactive **Dash** dashboards with real-time **Databricks** integration to visualize risk metrics and model outputs for senior leadership.

PROJECTS

Machine Learning Portfolio Optimizer | *Python, Flask, XGBoost, React, OpenAI API* July 2025 – Present

- Shipped a **Flask microservice + React dashboard** that runs live **Markowitz optimizations** on Yahoo Finance data, enforcing top-*k* portfolio constraints and surfacing **allocation diagnostics** in real time.
- Engineered numerous **time-series features** (technical, momentum, risk) and tuned an **XGBoost classifier** for **3-month S&P outperformance** across 80+ tech equities, layering **cross-validation**, **rolling holdout evaluation**, and **logistic/random-forest baselines** to monitor accuracy, **ROC-AUC**, and calibration.
- Built an **NLP pipeline** combining Finnhub feeds with **OpenAI GPT-4 sentiment analysis**, **prompt-engineering** transformer outputs into structured polarity scores, merging Google Trends alternative data, and packaging the signals for downstream **machine learning** and **ensemble modeling**.

EDUCATION

University of Virginia Charlottesville, VA
Bachelor of Arts in Computer Science and Applied Statistics; GPA: 3.74 Aug 2022 – May 2026

- Relevant Coursework:** Computation with Python, Data Structures and Algorithms, Object-Oriented Programming, Machine Learning, Computer Architecture, Software Development Essentials, Discrete Mathematics, Web-Based Applications, Cybersecurity, Linear Models